

Analysis plan and deviations

Hypotheses, inclusion rules and decision criteria fixed before the confirmatory dataset existed, followed by thirteen itemized deviations from that plan.

Supporting information for *Street slope and reported property crime in nine United States cities: terrain measurement, target exposure, and tests of candidate mechanisms*.

Topography and property crime: testing the offender-effort mechanism

Written 2026-08-04, **before** the confirmatory dataset (street segments, building-based exposure, expanded city panel) exists. Analyses run prior to this date are labelled EXPLORATORY throughout and are reported as such. This document fixes the confirmatory tests, the inclusion rules, and the decision criteria in advance.

Intended registry: OSF. Format follows AsPredicted's nine questions, expanded.

1. Background and what is already known

Two published studies report that property crime is lower at higher elevation:

- Breetzke (2012), *Applied Geography* 34:66–75. Tshwane, South Africa. Burglary 2003–2006.

Higher altitude → lower burglary risk. **Slope had no effect.**

- Kim & Wo (2023), *Journal of Urban Affairs* 45(6):1120–1144. San Francisco street segments.

Elevation *differences within the surrounding ¼ mile* reduced crime risk more than a segment's own elevation or slope. Network betweenness increased risk.

Both attribute the association to **offender effort** — climbing costs energy, so offenders substitute toward flatter targets — and **neither tests that attribution**. No multi-city study exists. No study in this literature has attempted to discriminate between candidate mechanisms.

2. Hypotheses

We treat the *existence* of a terrain–crime association as established and test the **mechanism**. Three candidates make different predictions:

	Mechanism	Prediction
M1	Effort — climbing and hauling cost energy	Deterrence scales with the physical work of the crime, especially the mass removed
M2	Exposure/risk — hills carry less through-traffic, have fewer escape routes, and are overlooked	Deterrence is roughly equal across crime types, and is mediated by network and visibility measures
M3	Confounded affluence — hills are wealthy	Association vanishes under within-neighbourhood comparison and tracks the city's elevation–income gradient

Primary hypothesis (H1, the no-loot control). If M1 holds, crimes in which nothing is carried away — vandalism and arson — must be substantially less deterred by relative height than theft, because their removal cost is zero by construction while their approach cost is unchanged.

H1: the terrain coefficient for no-loot crime is *closer to zero* than the terrain coefficient for theft, in most cities.

We predict H1 will be FALSIFIED — that the two coefficients will be statistically indistinguishable. Exploratory work in four cities is consistent with that, and this registration exists to test it confirmatorily on new units, new denominators, and new cities.

H2 (loot-mass gradient). Under M1, the terrain coefficient varies monotonically with the mass removed, across the ordered ladder pocketable → light → medium → heavy → very heavy. Predicted: no monotone trend.

H3 (mediation). Under M2, controlling for network betweenness, permeability, egress count, and the visibility proxy substantially attenuates the terrain coefficient. Predicted: substantial attenuation (see §7 for the threshold).

H4 (motor vehicle theft). MVT's "loot" self-propels, so its removal cost is independent of gradient. Under M1 it should be among the least deterred; under M2 it should be among the most deterred, since hilltop streets are conspicuous with few exits.

H5 (relief floor / placebo). Cities below the relief floor (§4) show terrain coefficients indistinguishable from zero. A non-zero effect in a flat city indicates residual confounding rather than terrain.

3. Design

Observational. Unit of analysis: **street segment** (block face between intersections),

matching Kim & Wo. Grid cells at 100 m are retained only as a robustness check.

Identification is **within census block group** throughout: block-group fixed effects are absorbed, so every comparison is between segments in the same neighbourhood — the top of a hill against the bottom of the same hill, same demographics, same police beat.

4. Inclusion criteria (fixed now, applied before outcome analysis)

A city enters the confirmatory panel only if:

1. **TPI SD ≥ 4 m** at the 500 m radius. Below this, a bare-earth DEM is dominated by levees, highway embankments, and fill rather than terrain. *(Motivating example: Baton Rouge spans 15.9 m of relief, 1 SD of TPI = 0.98 m, and returns +22.9%, $z = 3.1$.)*
2. **$\geq 20,000$ classified property-crime incidents** since 2018-01-01.
3. **≥ 1.0 incident per analysis unit**, ensuring the crime feed actually covers the geography assigned to it. *(Marin County: 1,313 km², 99% of cells empty, because the Sheriff polices only unincorporated areas.)*
4. **Point-level geocoding**, not block-face or centroid-snapped only.
5. For H1 specifically: **$\geq 3,000$ no-loot incidents** (vandalism, criminal damage, arson). Cities are screened on these criteria using terrain and volume only — **never on the outcome coefficient.**

5. Variables

Exposure (independent). Relative height = Topographic Position Index, elevation minus the mean elevation within radius R , from USGS 3DEP 10 m DEM. Primary $R = 500$ m; the full sweep $R \in \{50, 100, 250, 500, 1000, 2000\}$ is reported. Standardised within city.

Outcome. Counts of property crime per segment since 2018-01-01, split into: MASS_1 pocketable, MASS_2 light, MASS_3 medium, MASS_4 heavy, MASS_5 very heavy/tools, MVT, NO_LOOT. The loot-mass ordering is fixed in `src/crime_classes.py` and **was set before any outcome analysis**.

Offset. Building-footprint-apportioned housing units and population. Area-apportioned exposure is a robustness check, not the primary.

Controls. Median household income, median home value, owner-occupancy share, vacancy rate, log exposure density; block-group fixed effects.

Mediators (H3). Betweenness, intersection density, permeability, egress count, dead-end status, stairs within 100 m, walk/drive network ratio, visibility proxy.

6. Statistical model

Poisson pseudo-maximum-likelihood with a log-exposure offset and absorbed block-group fixed effects; standard errors clustered on block group. Poisson rather than OLS on rates because the outcome is an overdispersed count with many zeros; PPML rather than dummy variables because ~500–2,500 fixed effects make IRLS diverge.

Spatial autocorrelation: primary specification uses cluster-robust SEs; a conditional autoregressive (BYM) specification is reported as the sensitivity analysis. Moran's I on residuals is reported for every model.

7. Decision rules (set in advance)

- **H1 falsified** if the 95% CI for the *difference* between the theft and no-loot terrain coefficients contains zero in $\geq$ half the qualifying cities, with a pooled difference whose CI contains zero. Difference tested by seemingly-unrelated estimation with city-clustered bootstrap (2,000 draws).
- **H2 falsified** if the inverse-variance-weighted slope of the coefficient on loot mass has a 95% CI containing zero.
- **H3 supported** if adding mediators attenuates the pooled terrain coefficient by $\geq 40\%$.
- **H4 resolved** by whether MVT ranks in the upper or lower half of deterrence.
- **H5 supported** if the pooled coefficient among sub-floor cities has a CI containing zero.
- **Substantive-size floor:** any effect smaller than **5% per SD** is reported as substantively negligible regardless of p-value. With tens of thousands of units, significance is cheap and this floor keeps the discussion honest.

8. Sample size and power

Determined by data availability, not by choice. Expected 6–12 qualifying cities and 15,000–30,000 segments per large city. We do not stop collection based on results, and we do not add or drop cities after seeing coefficients. The panel is frozen once §4 is applied.

9. What would change our conclusion

Stated explicitly to preclude post-hoc reinterpretation:

- If no-loot crime is clearly **less** deterred than theft (pooled difference CI excluding zero, in the predicted direction), **M1 survives and our thesis is wrong**.
- If mediators do not attenuate the terrain coefficient, **M2 is not the answer either**, and the association is left unexplained — which we will report as such rather than fitting a third story to the residual.
- If the terrain coefficient does not survive block-group fixed effects and building-based exposure, **M3 wins** and the published finding is confounding.

10. Known limitations, acknowledged in advance

1. **Reported crime only.** Reporting rates rise with income, which correlates with elevation. This biases *against* finding a negative terrain effect.
2. **Crime-type harmonisation across cities is imperfect.** Departments use different taxonomies; the text classifier in `src/crime_classes.py` is coarse in cities that publish only a top-level offense category.
3. **Bare-earth DEMs still contain built terrain** — embankments, retaining walls, cuts. This motivates the relief floor but does not eliminate the problem.
4. **No offender residence data**, so crime-location-choice models (Bernasco) are out of reach and the substitution radius is inferred rather than observed.
5. **Directional movement costs are not identifiable from a raster.** Approach and loaded-escape cost are both functions of the same gradient field ($r = -0.80$ with TPI in San Francisco). Separating them requires an asymmetric network. We report this as a methodological finding rather than attempting the decomposition.
6. **Ecological inference.** Conclusions concern places, not people.

11. Exploratory work already conducted (full disclosure)

Conducted before this registration and reported as exploratory:

San Francisco pilot at 100 m cells (attenuation path, radius sweep, loot-mass ladder); an 11-city panel with area-apportioned exposure; a two-stage decomposition of city-level coefficients on the elevation–income correlation; and a directional round-trip cost model of our own predicting sign reversal with loot mass, which was **not supported** (slope $-0.0018/\text{kg}$, CI $[-0.010, +0.018]$). All p-values from that work are descriptive. The confirmatory tests above are run on new units, new denominators, and an expanded panel.

Addendum — deviations from this pre-registration

Recorded 2026-08-04, after the confirmatory analysis. Listed so that readers can discount them appropriately rather than discover them.

D1. The treatment variable changed, and this was not anticipated

This document registers **relative height (TPI)** as the exposure, following the published literature. The confirmatory analysis found relative height to be inconsistent in sign across cities, while **slope** was strongly and consistently negative everywhere. The headline analysis therefore reports slope.

This was **not** pre-specified and must be read as exploratory. In mitigation: slope was always in the variable set as a control, it was not selected from a large pool (there are two candidate terrain measures, not twenty), and the selection was made on *consistency across cities*, not on statistical significance in a single one. The registered hypotheses H1–H4 were then re-run on slope and are reported for both variables in the manuscript.

D2. A slope floor was added post hoc

Section 4 registers a relief floor of TPI SD ≥ 4 m. Applying the analysis to slope revealed the same pathology in a different variable: cities with almost no gradient (Chicago, mean slope 0.96°) return implausibly large per-degree effects. We therefore added a **slope floor of within-city slope SD $\geq 3^\circ$ **, which retains San Francisco, Seattle, and Cincinnati.

This threshold was chosen **after seeing the estimates** and is therefore post hoc. Two things argue it is not merely a device for producing a clean result:

1. It is the direct analogue of a criterion registered in advance for the other terrain variable, applied for the identical reason.
2. It is validated by a statistic that was not used to choose it: heterogeneity. Above the floor, $I^2 = 0.00$ ($Q = 1.28$, 2 df). Below it, $I^2 = 0.88$. A threshold chosen to flatter a result would not be expected to produce perfect agreement on one side and violent disagreement on the other.

Readers who reject the floor should read the six-city result: the per-degree effect remains negative and significant in every city, but ranges from -5.7% to -21.6% and is decisively heterogeneous.

D3. Mediator set reduced

The registered mediator list included a visibility proxy. On construction it proved near-collinear with relative height ($r = 0.80$) — it is close to a monotone transform of the treatment rather than a mediator of it — and was dropped. Testing the visibility channel requires building heights, not a bare-earth surface. Reported in the manuscript. Additionally, betweenness is computed with **travel-time** rather than distance weighting. On San Francisco's network, distance-weighted betweenness is confounded with the treatment: its top-ranked streets are residential lanes over the Twin Peaks ridge, because terrain forces cross-town paths through them. The two orderings correlate at only $\rho = 0.68$.

D4. Panel smaller than anticipated

Section 8 anticipated 6–12 qualifying cities. After the slope floor, three cities carry the headline estimate and six carry the no-loot control. This is the principal limitation of the study and is stated as such.

D5. Out-of-sample test, and a claim withdrawn

After the addendum above was written, three further cities (Pittsburgh, Baltimore, Charlotte) were harvested from ArcGIS sources. They played no part in selecting slope as the treatment or in setting the 3° gradient floor, so they function as an out-of-sample test of both. Two results:

The gradient floor is confirmed. Baltimore (2.86°) and Charlotte (2.10°) fall below it and return −14.1% and −20.7% per degree, squarely in the inflated band occupied by the other sub-floor cities and far outside the above-floor range. This was predicted before the models were run.

The homogeneity claim is withdrawn. With three qualifying cities the pooled per-degree effect showed $I^2 = 0.00$ and an earlier draft described it as a constant. Adding Pittsburgh moves this to $I^2 = 0.86$ ($Q = 21.1$ on 3 df, $p = 1 \times 10^{-4}$). The original figure was an artifact of a Q test with two degrees of freedom, which has almost no power — a limitation we had flagged for the H1 test and failed to apply to the headline. The manuscript now reports the pooled estimate with heterogeneity attached and explicitly warns against treating it as transportable.

We record this because it is the clearest illustration in the project of why the $k = 3$ caveat mattered, and because the corrected claim is weaker than the one it replaces.

D6. Positioning corrected

An earlier draft framed the slope finding as novel and as reversing Breetzke (2012). This was wrong. **Haberman & Kelsay (2021, *Journal of Quantitative Criminology*)** had already established a slope–crime association for robbery in Cincinnati, at a magnitude (~7.9% per degree) close to ours. The contribution is therefore generalisation from robbery to property crime across cities, plus the mechanism tests — which address the three explanations Haberman and Kelsay explicitly named and left untested. The manuscript has been repositioned accordingly.

D7. Defects found by audit, and what they cost

Four coding defects were found by validation rather than by inspection. All were silent — totals looked plausible, models converged, coefficients were stable — and all are recorded here because each one changed which analyses were possible.

1. `Series.astype(str)` **preserves NaN on pandas 3.0.5**. Both harvesters concatenate up to three description columns; a single null column turned the whole string to NaN and sent the row to the unclassified bucket. This emptied every non-theft class in Cincinnati.
2. **NIBRS 23G/24I parts and plate thefts matched the motor-vehicle-theft rule**. In Seattle that merged 32,501 parts thefts into 50,824 real vehicle thefts — a 49% inflation of the one class where the candidate mechanisms make opposite predictions — while leaving the heaviest loot rung empty.
3. **"criminal damage" does not match Ohio's "CRIMINAL DAMAGING/ENDANGERING."** Cincinnati's 28,513 no-loot records fell to the unclassified bucket, and the city was excluded from this study's primary test for apparently lacking vandalism data.
4. **Cincinnati's registry row read only** offense, a UCR top-level field, omitting the NIBRS sub-code the loot ladder needs. Without it the city had zero MVT, MASS_1, MASS_2 and MASS_5.

All four are fixed and the panel was rebuilt. The headline moved from –6.61% to –6.61%; the primary test gained a city.

Process note, recorded against ourselves. The panel rebuild was started while the classifier was still being edited, and Cincinnati was consequently built against a half-applied rule set. It was caught by comparing class counts before and after (outputs/classifier_fix_impact.csv) and rebuilt. Had that comparison not been run, the

error would have shipped.

D8. Los Angeles recovered; two further classifier gaps closed

Los Angeles had been absent from the analysis. The cause was a field-selection defect, not a data limitation: the registry ranked description columns in dataset order, and Los Angeles lists `weapon_desc`, `premis_desc` and `vict_desc` before `crm_cd_desc`. Since the harvester concatenates the first three, the classifier was reading a weapon, a premise and a victim's ethnicity while the only field naming the offense was discarded. `registry.py` now *ranks* description columns, promoting offense-like names and demoting victim, weapon, premises and case-status fields.

Recovering the city exposed two further gaps in the classifier, both fixed:

- * **Inverted motor-vehicle-theft wording.** Los Angeles publishes vehicle theft as "VEHICLE - STOLEN" (115,184 records) and Prince George's County as "AUTO, STOLEN". Neither contains "stolen vehicle" in that order, so both fell to the unclassified bucket and the city lost its MVT class entirely.
 - * **Vehicle burglary read as building burglary.** Los Angeles records theft from a car as "BURGLARY FROM VEHICLE" (63,515 records) and Nashville as "BURGLARY - MOTOR VEHICLE". The generic burglary rule fired first, moving a rung-2 offense to rung 4. The vehicle-burglary pattern is now tested ahead of both the burglary and motor-vehicle-theft rules.
- A third repair followed: separator normalisation collapses the slash in "THEFT F/AUTO", so the literal pattern `f/auto` stopped matching and a theft-from-vehicle fell to the generic rung. The pattern is now `\bf[\\s/]*auto\b`.

Impact on existing results was checked before each change was adopted, not after. Every new pattern was run against the description vocabulary of all built cities. They match **nothing** in San Francisco, Seattle, Cincinnati, Kansas City, Montgomery County, Pittsburgh or Los Angeles, and 9,036 records in Chicago. All cities above the gradient floor are therefore unaffected and **the headline estimate does not change**; only Chicago, which is below the floor, was rebuilt (−19.76% to −19.76% per degree).

Re-scored against the hand-coded validation sample using the pipeline's own normalisation, the repairs fix four strings and break none: primary-sample accuracy rises from 0.922 to 0.932 string-level and 0.968 to 0.974 incident-weighted, the holdout is unchanged, and NO_LOOT precision and recall are untouched at 0.956 and 1.000. The holdout being flat is the

honest reading — these repairs help specific cities rather than raising accuracy in general.

D9. Prior-estimate conversion corrected

An earlier draft converted Haberman and Kelsay's coefficient to -7.9% per degree using a linear shortcut ($4.5 \div 0.573$). Under a log link the conversion compounds:

$\beta = \ln(0.955) \times \tan(1^\circ) \times 100 = -0.0804$, giving **-7.72% per degree**. The manuscript now carries the compounded figure.

We were unable to obtain the full text of Haberman and Kelsay — the journal is paywalled and the repository copy returns HTTP 403 — so the effect size, the percent-grade units and the three proposed mechanisms are drawn from the published abstract and secondary summaries. This is flagged in the manuscript and should be confirmed against the original before submission.

D10. The gradient floor was demoted from a sample rule to a sensitivity analysis

The plan set no terrain threshold. D2 recorded that a 3° within-city slope standard deviation floor was added after seeing terrain data, and defended it on measurement grounds. The defence is still the one I would give, but the objection stands: a threshold chosen after inspecting the data is a researcher degree of freedom no matter how good the argument attached to it, and this one set the headline number.

The primary analysis is now a random-effects meta-regression of the city coefficient on that city's slope standard deviation, centred at 3° , using all nine cities. The moderator is significant ($+4.06\%$ per degree of slope SD, 95% CI $[+0.84, +7.38]$, $p = 0.020$) and accounts for 59% of the between-city variance. The threshold split is retained below it as a sensitivity analysis. This says everything the floor said, uses every city, and does not require anything to be cut.

The pooling method also changed. The plan and every draft through the first submission version used a fixed-effect inverse-variance pool, which assumes a common effect and narrows as cities are added even when they disagree. The pools are now random effects with REML τ^2 and a Hartung–Knapp interval. For the four higher-gradient cities this moves the headline from -6.61% $[-7.39, -5.82]$ to **-6.89% $[-9.51, -4.21]$** , and the prediction interval for a new city is $[-13.71, +0.46]$. The point estimate barely moved; the honest uncertainty around it roughly doubled.

D11. The no-loot test was re-specified after external review

Three changes, all of which make the test harder to pass rather than easier.

Equivalence testing replaced the null. The plan's decision rule was "indistinguishable in at least half the cities plus a pooled interval spanning zero." A null result is also what an underpowered study produces, so that rule cannot support the claim it was being used for. The contrast is now evaluated by two one-sided tests against a margin of half the headline effect (0.0342 on the log scale, 3.48 pp per degree), fixed before the tests were run. Pooled difference +0.55 pp/degree, TOST $p < 10^{-10}$.

Arson was separated from vandalism. Arson involves preparation and acute escape risk; grouping it with graffiti made the control less homogeneous than the paper implied. It is 1.6–3.9% of no-loot incidents in three cities and 0.02% in Cincinnati. Dropping it moves the pooled difference from +0.55 to **+0.09 pp/degree** [−2.28, +2.52].

The two offense groups were matched on time. Vandalism concentrates at night and theft does not. Reweighting no-loot incidents to theft's hour-of-day × day-of-week distribution gives **+0.10 pp/degree** [−2.25, +2.50].

Both variants required re-downloading the four higher-gradient cities with the incident timestamp and offense text retained, which the original harvest discarded (`src/harvest_dated.py`). I no longer describe the comparison as a placebo. It is a bounded falsification of a loaded-haul mechanism and nothing wider.

D12. Reporting added that the plan did not require

Estimator diagnostics (convergence, iterations, separation, singleton and all-zero fixed-effect groups, sample reduction); a 1 km spatial block bootstrap alongside the cluster-robust errors, which shows block-group clustering was mildly *anticonservative*, up to 1.30× in Seattle, reversing a claim made in an earlier draft; classification error propagated to the coefficients through 200 relabelling draws from the audit confusion matrix (it contributes 2–3% of the sampling standard error); pre- and post-March-2020 estimates; and the target-count denominator extended from San Francisco to all nine cities, where it gives a mixed result rather than the clean San Francisco one.

Two citation errors were also corrected: reference 4 conflated a San Francisco bicycle theft paper with a separate Toronto paper, and Herzog was dated 2013 rather than 2014.

The Ye and Becker multi-city elevation literature and the offender location-choice

literature were missed entirely in the original scan and are now cited.

D13. The multi-city target-denominator extension was retracted and re-run

The first version of D12's footprint-denominator extension reported estimates for all nine cities, including a -48.5% per degree figure for San Francisco and a -16.3% for Seattle. Those were not estimates. Five of the nine fits had stopped at the iteration cap with a maximum remaining score in the thousands and were returning standard errors around $1e-11$, which is a degenerate sandwich rather than a precise coefficient. The estimator's own convergence diagnostics — added in D12 — would have caught it, but `target_multicity.py` never checked them.

Two defects, both fixed. The control `log_density` was being redefined as the log of whichever denominator was in play, so the control set changed along with the offset and became the log of the offset itself; it is now always residential density on the original housing exposure, and only the offset changes. And convergence is now judged on the first-order condition (max absolute score, scaled by total events) rather than on a coefficient-change flag, with the iteration cap raised from 60 to 200.

After the fix, five cities (Charlotte, Seattle, Cincinnati, Chicago, Montgomery County) still fail to converge and are not reported. In the four that do, the denominator shifts the estimate by $+0.99$, -1.49 , -0.57 and $+0.17$ percentage points — two each way, all small. The extension neither supports nor contradicts the San Francisco result; it is uninformative, and the manuscript now says so.

The iteration-cap change is inert for every other model in the paper: all nine headline coefficients are identical to machine precision before and after, since they converge in eight to eleven iterations.